

Matt Brown

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Summary

Innovative and user-focused Software Engineer with over 15 years of experience designing, developing, and maintaining scalable software solutions. Committed to writing clean, reusable code and continuously improving systems to meet evolving user and business needs.

Skills

TypeScript, JavaScript, Python, Java, React, Node.js, D3.js, ROS2, Git, Azure, Jenkins, Docker, Kubernetes, Elasticsearch, MySQL, Agile Development, DevSecOps, Project Management, Personnel Management, Mentorship

Experience

Scientist, Software Engineer | 08/2025 - present | L3 Harris

- Engineered ROS2 nodes from scratch using Python, standing up new microservices to power embedded software components, enabling replacement of a core piece of software previously supplied by a competitor.
- Architected automated build pipelines in Jenkins for over 20 different software components, crafting individualized Docker containers for the unique build environments, ultimately slashing build times from 30+ minutes to 1-2 minutes.
- Standardized and documented build steps across the DevSecOps pipelines, configuring Jenkins, Bitbucket, and Artifactory so team members deploying the software components could quickly locate and access the latest versions.
- Built a framework to simulate software components in Azure Kubernetes pods, enabling integration testing between components before deployment to actual hardware, thereby reducing the risk of deployment issues during hardware testing or demo preparation.

Software Engineer | 03/2022 - 01/2025 | Microsoft Cloud+AI Platform | Elkridge, MD

- Streamlined the user experience for initializing machine learning compute resources, harnessing TypeScript and React to create reusable components, significantly improving the user's interactions with the application.
- Prepared Microsoft Azure's new AI Foundry product for initial release at Microsoft Ignite, turning Figma designs into working features utilizing Fluent UI and React components, having new features available for live demonstration and public launch at the Ignite conference.
- Drove the efforts to find and fix bugs with ML Platform and AI Foundry deployments to sovereign clouds, updating code while tracking and assigning bugs, allowing the sovereign clouds to be available to users within contractual deadlines.
- Transformed Azure's ML Studio application, iterating on designs with the creative team and modernizing the codebase, addressing actual user feedback, and providing a better overall user experience.
- Updated the Azure ML CLI and Python SDK to work in the air-gapped environment, fixing bugs and establishing an automated Azure DevOps CI/CD pipeline, enabling users in these environments to access and employ the tools.

Technical Task Order Lead | 08/2020 - 03/2022 | Northrop Grumman/Peraton | Annapolis Junction, MD

- Supervised a task order consisting of 21 engineers across three different teams, communicating status and progress to customer and program leadership while keeping the team fully staffed and working efficiently, gaining customer praise and recognition in the form of 99% award fees.
- Refocused a team of 14 developers after a span of poor agile leadership, becoming the scrum master and implementing agile practices, facilitating continuous improvement, and keeping the project 100% staffed.

Engineering Manager | 06/2020 - 03/2022 | Northrop Grumman/Peraton | Annapolis Junction, MD

- Guided a group of 13 engineers across all skill levels, giving career coaching, submitting awards, handling promotions, and managing performance reviews, resulting in three promotions and over 90% retention during a difficult post-acquisition transition.

Technical Project Manager/ Lead | 01/2018 - 08/2020 | Northrop Grumman | Annapolis Junction, MD

- Enabled a team of developers producing extensions to an existing web application, coordinating with customers and business stakeholders, and prioritizing courses of action, optimizing business impact from our work
- Executed multiple customer engagement sessions, participating in cycles of rapid application prototyping with Java and Kafka, and guiding our research and development efforts based on customer feedback, thus building strong customer relationships and trust in our team's abilities.
- Organized summer intern mentor program, pairing mentors with 100+ interns across 6 geographic areas and collecting feedback on the program, paving the way for effective interns who had positive encounters.

Cyber Software Engineer | 05/2014 - 12/2017 | Northrop Grumman | Annapolis Junction, MD

- Engineered a new application for monitoring system status, utilizing JavaScript, React, Java, D3.js, and Elasticsearch, delivering the initial solution featuring real-time status and replacing an outdated system.
- Devised Agile strategy, implementing team retrospectives and using them to move the team towards a more agile workflow, empowering team members to operate efficiently and be more engaged in their work.

Software Engineer | 04/2010 - 04/2014 | University of Maryland CATT Lab | College Park, MD

- Introduced a new version of an application, trimming down the PHP codebase into a lightweight Python/Django app, permitting the initial public version of a previously exclusive website.
- Analyzed ways to display historic, real-time, and predicted weather data, processing the data and solving issues like latency and aesthetics, ultimately supplying new insights to users.

Education

Master of Science in Computer Science | Johns Hopkins University | Baltimore, MD | 2021

Bachelor of Science in Computer Science | University of Maryland | College Park, MD